Measurement passport `pilot_gpt4o_mini_v34`

Totals

- points: 2958 published, 2958 answered
- tokens: prompt 7850106, response 8091766 (provider counters)
- money: **$6.0326** - cost of the published points (provider tokens × the tariff of the measurement)
- Bernoulli trials: 642430 in 2958 answered points - these are not model requests
- **provider counters are incomplete**: 6 billed points without counters - the token sums and the money are understated

Passport alarms (1):

- warning `INCOMPLETE_TOKEN_COUNTERS`: some billed points carry no provider token counters: the token sums and the money are understated, and the missing part cannot be recovered

Capacities and loads

vspmr_glmm_v1

Quantity	Value ± 1.96·SE	Unit	Status	Note
IEC, q = 0.95	36.58 ± 5.41 (binned) · 38.06 ± 6.13 (by law)	rules	measured · measured	how many rules the model holds above level 0.95; the gap between the two calculations measures the approximation
IEC, q = 0.5	43.90 ± 6.62 (binned) · 46.09 ± 6.61 (by law)	rules	measured · measured	how many rules the model holds above level 0.5; the gap between the two calculations measures the approximation
IGC, q = 0.95	18.69 ± 3.62 (binned) · 17.55 ± 4.04 (by law)	rules	measured · measured	how many rules the model holds above level 0.95 when every word of the text is replaced (p = 1); the gap between the two calculations measures the approximation
IGC, q = 0.5	29.51 ± 5.92 (binned) · 29.76 ± 5.11 (by law)	rules	measured · measured	how many rules the model holds above level 0.5 when every word of the text is replaced (p = 1); the gap between the two calculations measures the approximation
price of a position doubling	2.67 ± 0.310	logit/doubling	measured	logit lost per doubling of the word position
[illegible]	[illegible] 1838 (by law)	[illegible]	exp measured	[illegible] level 0.95 rules reserve; texts mea-

Figures

`vspmr_glmm_v1`

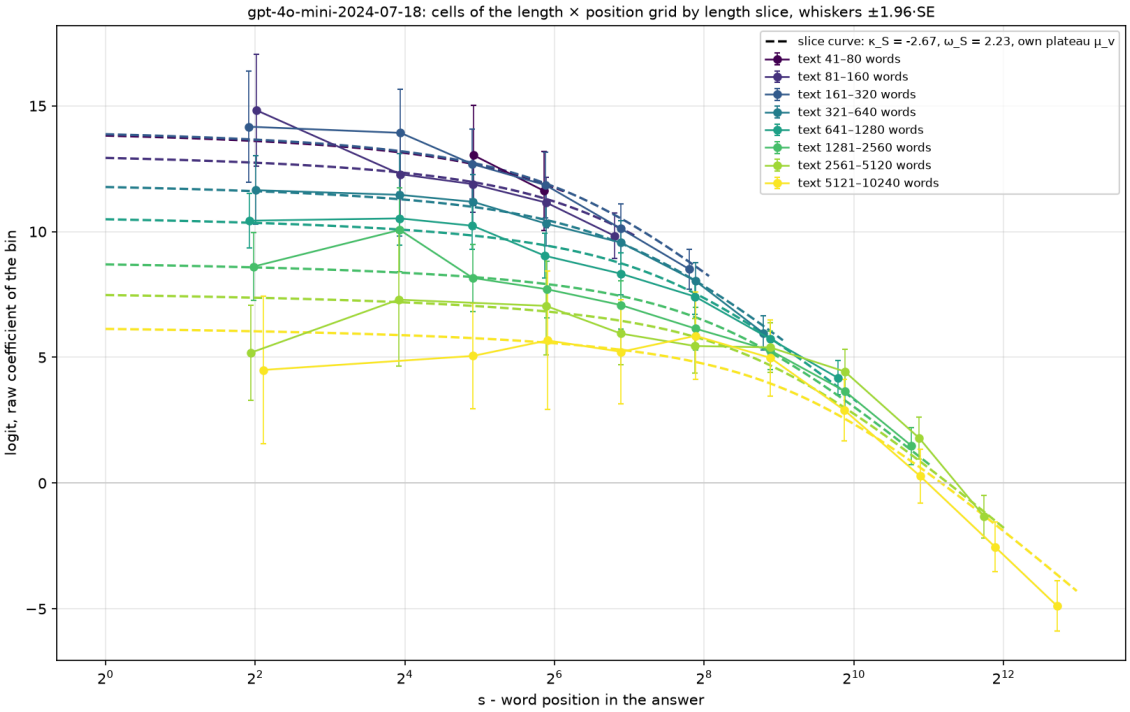


Figure 1: Grid cells by length slice and the slice curves, before the shift (raw coefficients)

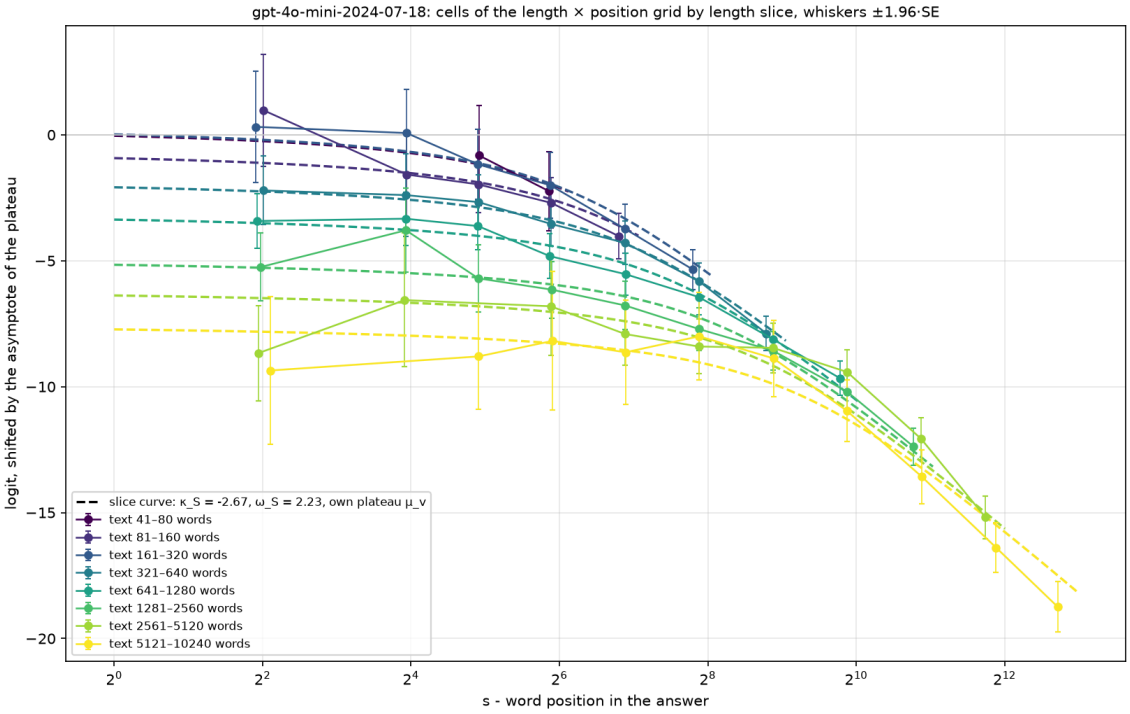


Figure 2: Grid cells by length slice and the slice curves, after the shift δ_VS

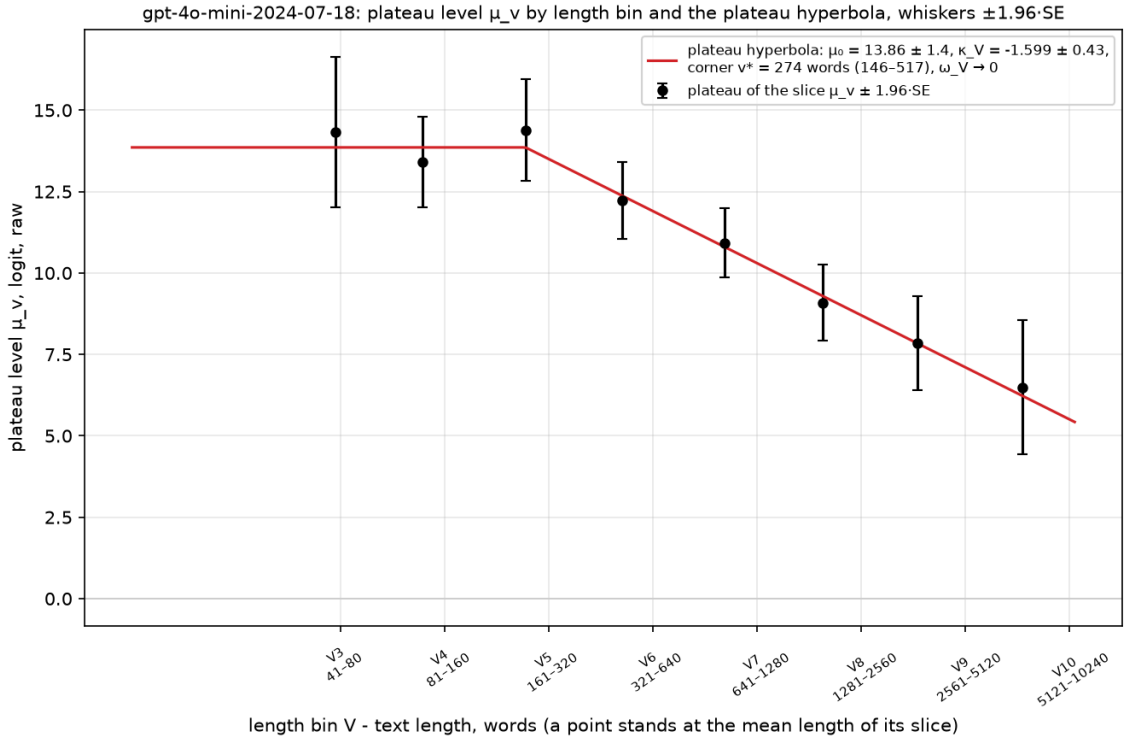


Figure 3: Plateau μ_v by length slice and the plateau hyperbola, before the shift

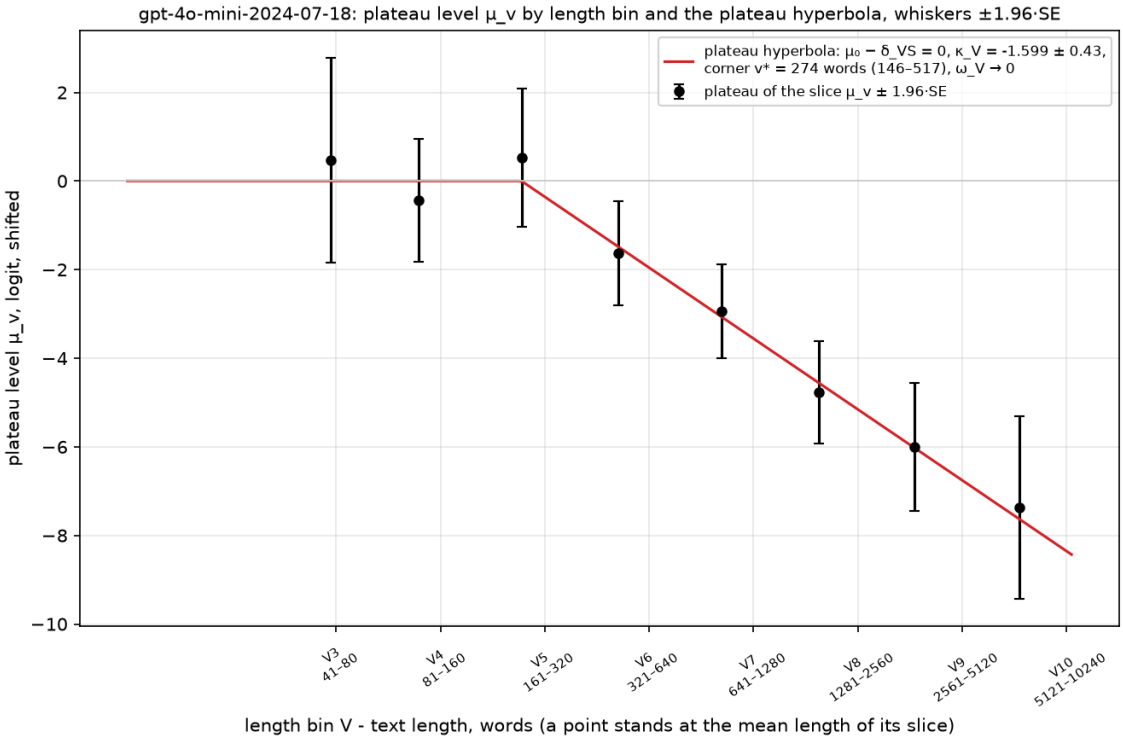


Figure 4: Plateau μ_v by length slice and the plateau hyperbola, after the shift δ_VS

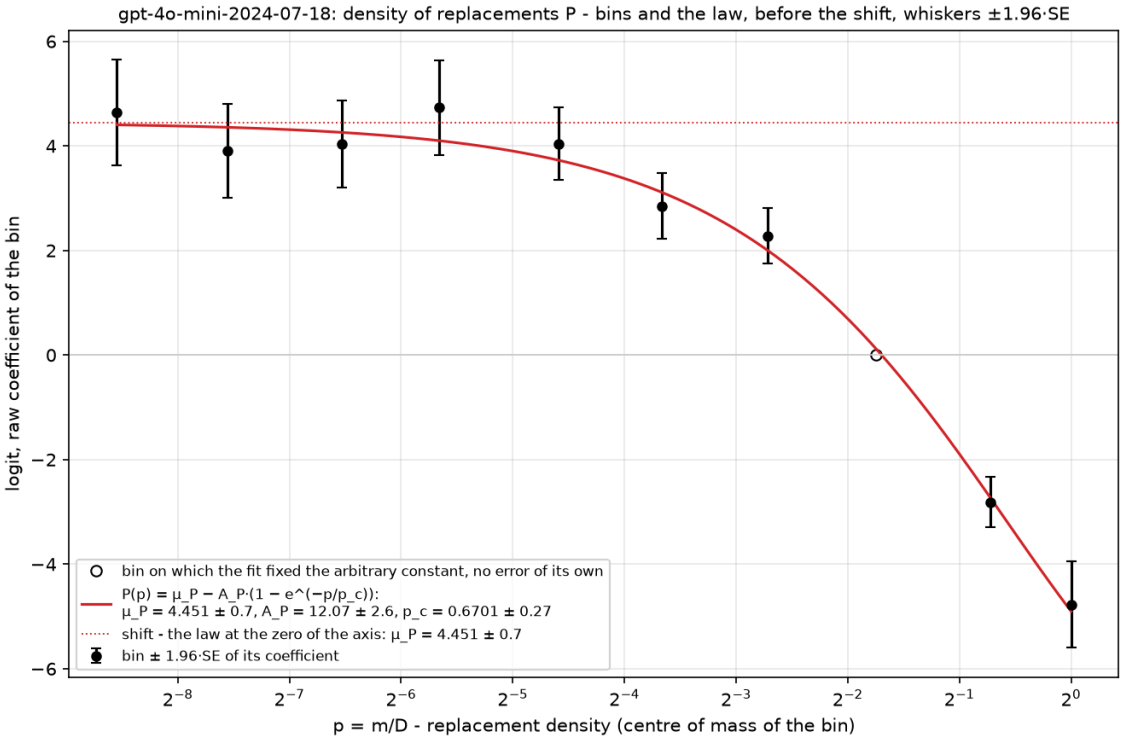


Figure 5: Density bins and their law, before the shift

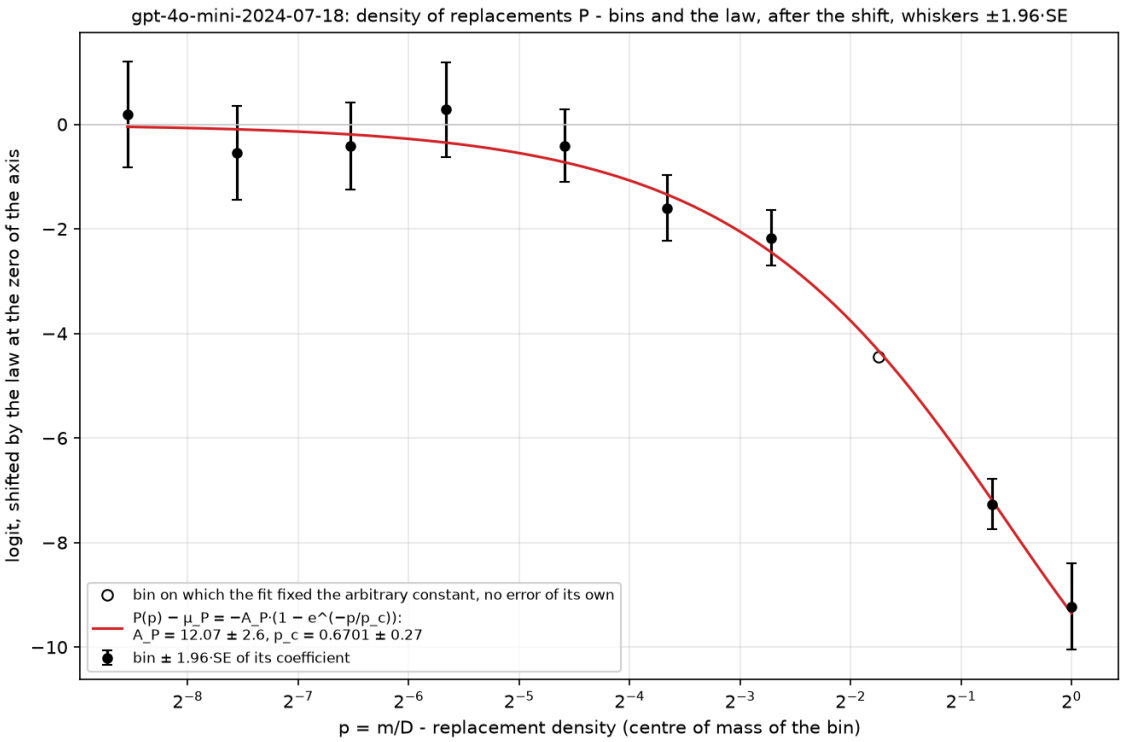


Figure 6: Density bins and their law, after the shift μ_P

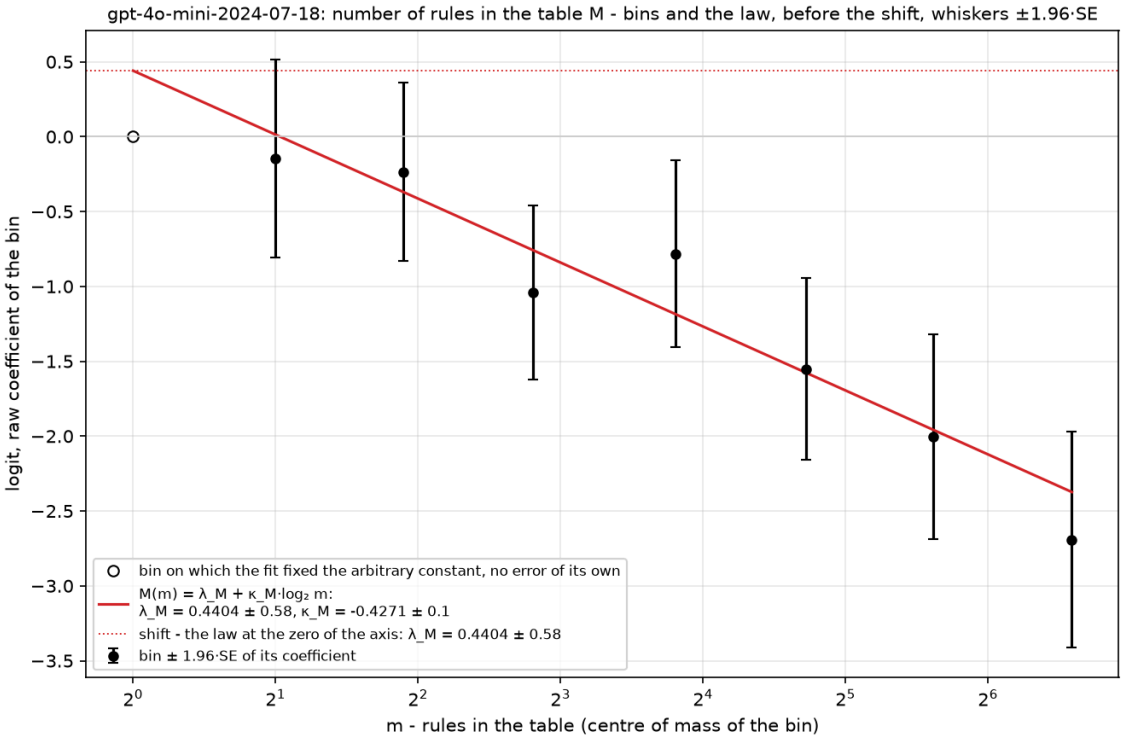


Figure 7: Rule-count bins and their law, before the shift

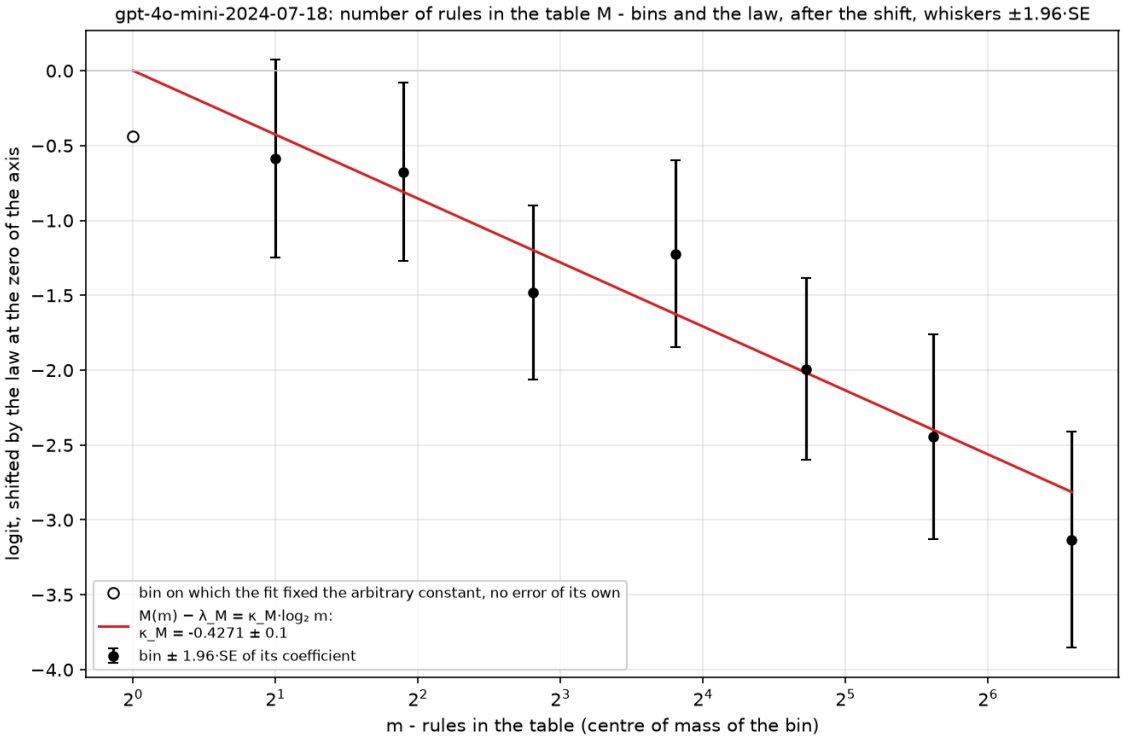


Figure 8: Rule-count bins and their law, after the shift λ_M

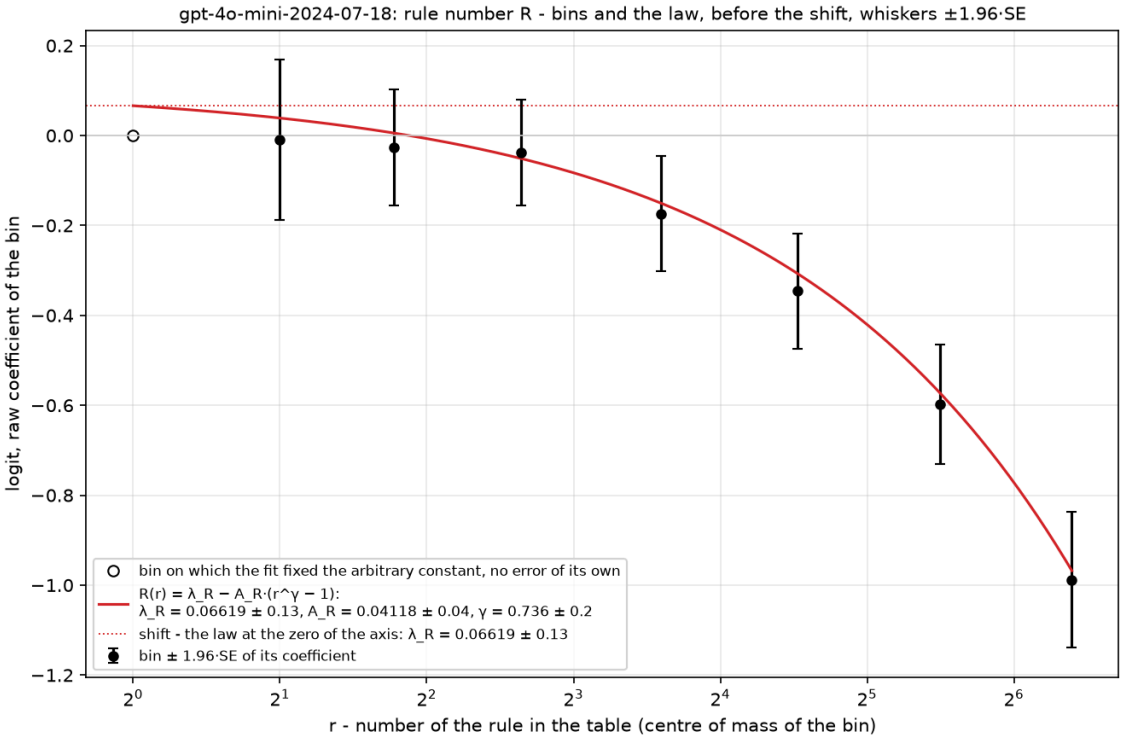


Figure 9: Rule-number bins and their law, before the shift

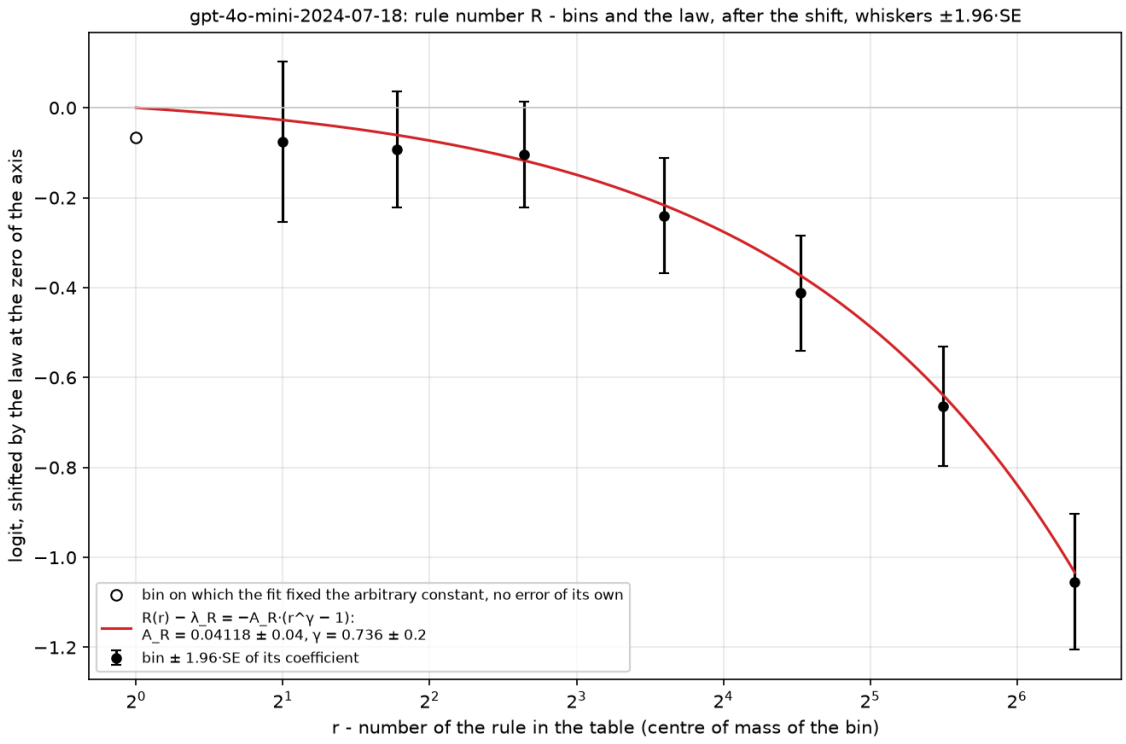


Figure 10: Rule-number bins and their law, after the shift λ_R

Model parameters

`vspmr_glmm_v1`

The passport model VS-PMR (`vspmr_glmm_v1`) gives the logit of a right replacement as

$$\eta = VS(v,s) + P(p) + M(m) + R(r) + b, \qquad b \sim N(0,\sigma^2)$$

b is the shift of a point, shared by all its trials and integrated out of the likelihood; the bins and laws describe the median point ($b = 0$). A capacity keeps its meaning - the share of right replacements over all points - so the level q is read at the threshold $\eta^*(q,\sigma^2)$ at which the median point gives the share q.

The second stage draws a law of its axis through the bins of every block:

$$VS = \frac{1}{2}\left[(\lambda_S + \kappa_S \ell) + \mu(v) - \sqrt{(\lambda_S + \kappa_S \ell - \mu(v))^2 + \kappa_S^2 \omega_S^2}\,\right], \qquad \ell = \log_2 s$$

$$\mu(v) = \mu_0 + \frac{\kappa_V}{2}\left[\log_2 \frac{v}{v_*} + \sqrt{\log_2^2 \frac{v}{v_*} + \omega_V^2}\,\right]$$

$$P(p) = \mu_P - A_P\,\left(\,1 - e^{-p/p_c}\,\right), \qquad M(m) = \lambda_M + \kappa_M \log_2 m, \qquad R(r) = \lambda_R - A_R\,(r^\gamma - 1)$$

Here s is the word position in the answer, $v = D \cdot n$ the text length in words, $p = m/D$ the replacement density, m the number of rules and r the rule number. The surface law is fitted in

two steps: every length slice has its own plateau μ_v with shared κ_S, λ_S, ω_S, and the plateau $\mu(v)$ is a hyperbola in the text length drawn through the μ_v: the horizontal asymptote μ_0 for short texts and the slope κ_V per doubling for long texts, with the corner at the length v_* and the width ω_V. The shift of a block is its law at the zero of its axis - for the surface the horizontal asymptote $\delta_{VS} = \mu_0$, approached at the start of the answer ($s \to 0$) of short texts ($v \to 0$); $p \to 0$, $m = 1$, $r = 1$ - and it goes into the level α_{phys}.

Status: `measured`. Zero point: surface: the horizontal asymptote μ_0 of its plateau - the start of the answer (s ⟶ 0) of short texts (v ⟶ 0); p ⟶ 0, m = 1, r = 1.

A law of its axis with a free shift is drawn through the bins of every block (generalised least squares by the covariance submatrix of the bins); the shift - the law at the zero of the axis - is subtracted from the bins and moved into the level. The prediction of the trials does not change. Values are ± 1.96·SE.

Quantity	Value
`α_phys`	18.81 ± 1.44
shift: length × position grid	13.86 ± 1.37
shift: replacement density P	4.45 ± 0.703
shift: number of rules M	0.440 ± 0.578
shift: rule number R	0.0662 ± 0.135

Block	Parameter	Value	Meaning
length × position grid	delta_VS	13.86 ± 1.37	shift of the grid: the horizontal asymptote μ_0 of the plateau (the start of the answer of short texts)
length × position grid	kappa_S	−2.67 ± 0.310	fatigue slope, logit per doubling of the word position
length × position grid	lambda_S	17.39 ± 3.37	level of the slanted fatigue asymptote at the first word (s = 1), relative to the shift delta_VS
length × position grid	omega_S	2.23 ± 0.528	transition width, doublings of the word position
length × position grid	kappa_V	−1.60 ± 0.434	slope of the slanted asymptote of the plateau: logit per doubling of a long text
length × position grid	v_star	274.18 ± 173.65	corner of the plateau: the text length in words where its asymptotes cross
length × position grid	omega_V	0.00100	width of the corner of the plateau, doublings of the text length; held on the bound of its range - not measured
length × position grid	χ^2 / points	56.41 / 56	—
replacement density P	mu_P	4.45 ± 0.703	density shift (the law at p ⟶ 0)
replacement density P	A_P	12.07 ± 2.63	loss at saturation of the density
replacement density P	p_c	0.670 ± 0.269	characteristic density
replacement density P	[illegible]	[illegible]	[illegible]

Fit diagnostics

`vspmr_glmm_v1`

- between-point variance $\hat{\sigma}^2$ (inside the fit): 5.90 ± 0.360
- within-cluster correlation $\hat{\rho}$: 0.642
- weighting by the cells of the run cube V × P × M: 212 cells, 2757 points, largest occupancy 62; effective points n_eff = 1148.89

Plateau of the surface by length slice

The grid law is fitted in two steps. Every length slice has its own plateau level `mu_v`, computed from its cells; the slope `kappa_S`, the level `lambda_S` of the slanted asymptote and the width `omega_S` are shared. The fit weights are the own variances of the cells, without their mutual dependence: the cells of a slice are strongly correlated, and the full covariance drove a plateau above all cells of its slice; the errors are carried by the full covariance. A slice enters the fit only if its plateau is computable from its points: at least two measured cells and at least one on the plateau (the plateau share is the largest share of the plateau in the derivative of a cell of the slice).

The plateau hyperbola $\mu(v) = \mu_0 + \frac{\kappa_V}{2}\left[\log_2 \frac{v}{v_*} + \sqrt{\log_2^2 \frac{v}{v_*} + \omega_V^2}\,\right]$ is drawn through the plateaus of all determined slices (at least five), weighted by their own variances, and its horizontal asymptote μ_0 - the plateau of short texts - is the shift `delta_VS` = 13.86 ± 1.37; `lambda_S`, `mu_v` and the hyperbola are published minus `delta_VS`.

Plateau hyperbola through the `mu_v` points: `kappa_V` = −1.60 ± 0.434, `v_star` = 274.18 ± 173.65 words, `omega_V` = 0.00100; held on the bound: `omega_V`; χ^2 = 1.28 on 8 points. Shared: `kappa_S` = −2.67 ± 0.310, `lambda_S` = 17.39 ± 3.37, `omega_S` = 2.23 ± 0.528; χ^2 by the own variances = 56.41 on 56 cells with 11 parameters.

Slice, words	ū	Cells	Plateau share	`mu_v`	Plateau of the grid law `μ(ū)`
41–80	1.95	2	0.767	0.474 ± 1.56	−0.000000219
81–160	2.79	5	0.951	−0.440 ± 0.616	−0.000000404
161–320	3.78	6	0.946	0.528 ± 1.22	−0.000598
321–640	4.70	7	0.958	−1.63 ± 1.32	−1.48
641–1280	5.69	8	0.965	−2.94 ± 1.39	−3.06
1281–2560	6.63	9	0.971	−4.77 ± 1.46	−4.56
2561–5120	7.54	9	0.975	−6.00 ± 1.62	−6.02
5121–10240	8.55	10	0.977	−7.37 ± 2.28	−7.63

Restored bins

length × position grid

Bin	Abscissa	Value	Law	Deviation, SE of the bin	Status
V1·S1 (words 1–20, positions 1–10)	s = 3.64, u = 0.00	—	—	—	excluded
V1·S2 (words 1–20, positions 11–20)	s = 14.40, u = 0.00	—	—	—	excluded
V2·S1 (words 21–40, positions 1–10)	s = 4.00, u = 1.00	—	—	—	excluded
V2·S2 (words 21–40, positions 11–20)	s = 15.45, u = 1.00	—	—	—	excluded
V2·S3 (words 21–40, positions 21–40)	s = 29.16, u = 1.00	—	—	—	excluded
V3·S1 (words 41–80, positions 1–10)	s = 4.01, u = 1.94	—	—	—	excluded
V3·S2 (words 41–80, positions 11–20)	s = 15.05, u = 1.93	—	—	—	excluded
V3·S3 (words 41–80, positions 21–40)	s = 30.19, u = 1.94	−0.819 ± 1.59	−1.54	+0.715	measured
V3·S4 (words 41–80, positions 41–80)	s = 58.01, u = 1.96	−2.24 ± 1.20	−2.24	+0.0113	measured
V4·S1 (words 81–160, positions 1–10)	s = 4.04, u = 2.76	0.972 ± 1.92	−0.701	+1.48	measured
V4·S2 (words 81–160, positions 11–20)	s = 15.36, u = 2.77	−1.58 ± 1.39	−1.12	−0.532	measured
V4·S3 (words 81–160, positions 21–40)	s = 30.03, u = 2.77	−1.97 ± 0.906	−1.54	−0.752	measured
V4·S4 (words 81–160, positions 41–80)	s = 59.26, u = 2.77	−2.71 ± 0.886	−2.28	−0.837	measured

replacement density P

Bin	Abscissa	Value	Law	Deviation, SE of the bin	Status
1 (p 0.00195–0.00391)	0.00268	0.188 ± 0.728	−0.0481	+0.459	measured
2 (p 0.00391–0.00781)	0.00531	−0.546 ± 0.578	−0.0952	−0.985	measured
3 (p 0.00781–0.0156)	0.0108	−0.416 ± 0.485	−0.194	−0.528	measured
4 (p 0.0156–0.0312)	0.0198	0.284 ± 0.645	−0.351	+1.37	measured
5 (p 0.0312–0.0625)	0.0416	−0.409 ± 0.390	−0.726	+0.901	measured
6 (p 0.0625–0.125)	0.0788	−1.61 ± 0.458	−1.34	−0.829	measured
7 (p 0.125–0.250)	0.152	−2.17 ± 0.569	−2.45	+1.02	measured
8 (p 0.250–0.500)	0.298	−4.45 ± 0.703	−4.33	—	measured
9 (p 0.500–1.00)	0.606	−7.26 ± 0.880	−7.18	−0.337	measured
10 (p 1.00–1.00)	1.00	−9.22 ± 1.04	−9.35	+0.306	measured

number of rules M

Bin	Abscissa	Value	Law	Deviation, SE of the bin	Status
1 (1–1)	1.00	−0.440 ± 0.578	0.00	—	measured
2 (2–2)	2.00	−0.586 ± 0.390	−0.427	−0.474	measured
3 (3–4)	3.73	−0.676 ± 0.347	−0.811	+0.443	measured
4 (5–8)	6.99	−1.48 ± 0.387	−1.20	−0.961	measured
5 (9–16)	13.97	−1.22 ± 0.507	−1.62	+1.26	measured
6 (17–32)	26.44	−1.99 ± 0.534	−2.02	+0.0843	measured
7 (33–64)	48.95	−2.45 ± 0.656	−2.40	−0.139	measured
8 (65–128)	96.20	−3.13 ± 0.730	−2.81	−0.863	measured

rule number R

Bin	Abscissa	Value	Law	Deviation, SE of the bin	Status
1 (1–1)	1.00	−0.0662 ± 0.135	0.00	—	measured
2 (2–2)	2.00	−0.0760 ± 0.127	−0.0274	−0.535	measured
3 (3–4)	3.43	−0.0923 ± 0.0600	−0.0609	−0.477	measured
4 (5–8)	6.23	−0.104 ± 0.0626	−0.117	+0.220	measured
5 (9–16)	12.07	−0.240 ± 0.103	−0.216	−0.363	measured
6 (17–32)	23.06	−0.412 ± 0.119	−0.374	−0.583	measured
7 (33–64)	45.08	−0.664 ± 0.127	−0.638	−0.387	measured
8 (65–128)	84.13	−1.05 ± 0.123	−1.03	−0.263	measured

Correlation of the components V, S, P, M, R

Canonical correlation of the increments of neighbouring bins of two components by the covariance of the fit: the largest correlation between a linear combination of the increments of one component and a combination of the increments of the other. One means that the fit cannot tell which of the two components explains an observation; a dash - the component has no increments.

Canonical correlation

	V	S	P	M	R
V	1.00	1.00	0.82	0.59	0.59
S	1.00	1.00	0.48	0.36	0.41
P	0.82	0.48	1.00	0.45	0.10
M	0.59	0.36	0.45	1.00	0.42
R	0.59	0.41	0.10	0.42	1.00

Collapse of the answer

995 of 2958 answered points lost the text - their answer ended in a failure of the third kind: lines dropped, merged or added in the middle, or an answer cut before the end of the text. Both characteristics of the collapse are described by the method of the passport model: the logit is split into the bins of its axes (the weighted logit, the cluster sandwich by points), a law is drawn through the bins of every axis, and the shift of an axis - its law at the zero of the axis - goes into the level.

Rate

The probability ψ that a point loses the text. Every answered point is one trial, the boundary cells of the area of analysis too, weighted by the run cube; the length law takes the inverse own variances of its bins as weights, the density law their full covariance:

$$\ln \frac{\psi}{1-\psi} = \alpha_\psi + V_\psi(\text{bin of } \ v) + P_\psi(\text{bin of } \ p)$$

$$V_\psi(v) = \lambda_{V\psi} + \kappa_\psi \log_2 \frac{v}{20}, \qquad P_\psi(p) = \mu_{P\psi} + A_\psi \, p^{\gamma_\psi}$$

$$\ln \frac{\psi}{1-\psi} = \lambda_\psi + \kappa_\psi \log_2 \frac{v}{20} + A_\psi \, p^{\gamma_\psi}, \qquad \lambda_\psi = \alpha_\psi + \lambda_{V\psi} + \mu_{P\psi}$$

- weighting by the cells of the run cube V × P × M: 274 cells, 2958 points, largest occupancy 62; effective points n_eff = 1259.94

Parameter	Value ± 1.96·SE	Note
λ_ψ	−20.00 ± 1.58	logit of the collapse at v = 20 words and p ⟶ 0
κ_ψ	2.40 ± 0.192	slope, logit per doubling of the text length
A_ψ	12.25 ± 1.01	density term at the full density p = 1
γ_ψ	0.706 ± 0.0931	exponent of the density term
λ_Vψ	−3.64 ± 2.12	length shift: the length law at v = 20
μ_Pψ	−5.33 ± 0.758	density shift: the density law at p ⟶ 0
χ^2 / points, `collapse_length`	2.12 / 7	the law against its bins
χ^2 / points, `collapse_density`	18.86 / 9	the law against its bins

The bins are fitted on 2917 trials (989 with the text lost); a bin whose trials all have one outcome has no estimate and goes out with its trials: `v_bin_1` (10 trials, all kept); `v_bin_2` (25 trials, all kept); `p_bin_1` (6 trials, all lost). The coefficients are raw logits relative to the reference bin of the axis (`0`), before the shift.

Axis	Bin	Value ± 1.96·SE
text length v, words	41–80	0.00
text length v, words	81–160	2.28 ± 2.05
text length v, words	161–320	6.18 ± 2.06
text length v, words	321–640	8.23 ± 2.14
text length v, words	641–1280	10.74 ± 2.16
text length v, words	1281–2560	13.39 ± 2.20
text length v, words	2561–5120	14.81 ± 2.21
text length v, words	5121–10240	16.82 ± 2.23
replacement density p	1/512–1/256	−5.05 ± 0.924
replacement density p	1/256–1/128	−4.04 ± 0.870
replacement density p	1/128–1/64	−5.00 ± 0.718
replacement density p	1/64–1/32	−4.96 ± 0.701
replacement density p	1/32–1/16	−3.74 ± 0.699
replacement density p	1/16–1/8	−3.37 ± 0.578
replacement density p	1/8–1/4	−2.03 ± 0.518
replacement density p	1/4–1/2	0.00
replacement density p	1/2–1	2.71 ± 0.534
replacement density p	1	7.04 ± 0.875

Check, the number of rules: the bins of the number of rules fitted together with the bins above; Wald statistic «all of them are zero» 9.78 with 7 degrees of freedom, p = 0.202; the largest change of a bin above 0.311 of its SE.

Number of rules	Value ± 1.96·SE
1	0.00
2	−0.124 ± 0.440
3–4	0.0583 ± 0.475
5–8	−0.466 ± 0.446
9–16	0.0933 ± 0.475
17–32	−0.293 ± 0.470
33–64	−0.293 ± 0.672
65–128	−0.0672 ± 0.629

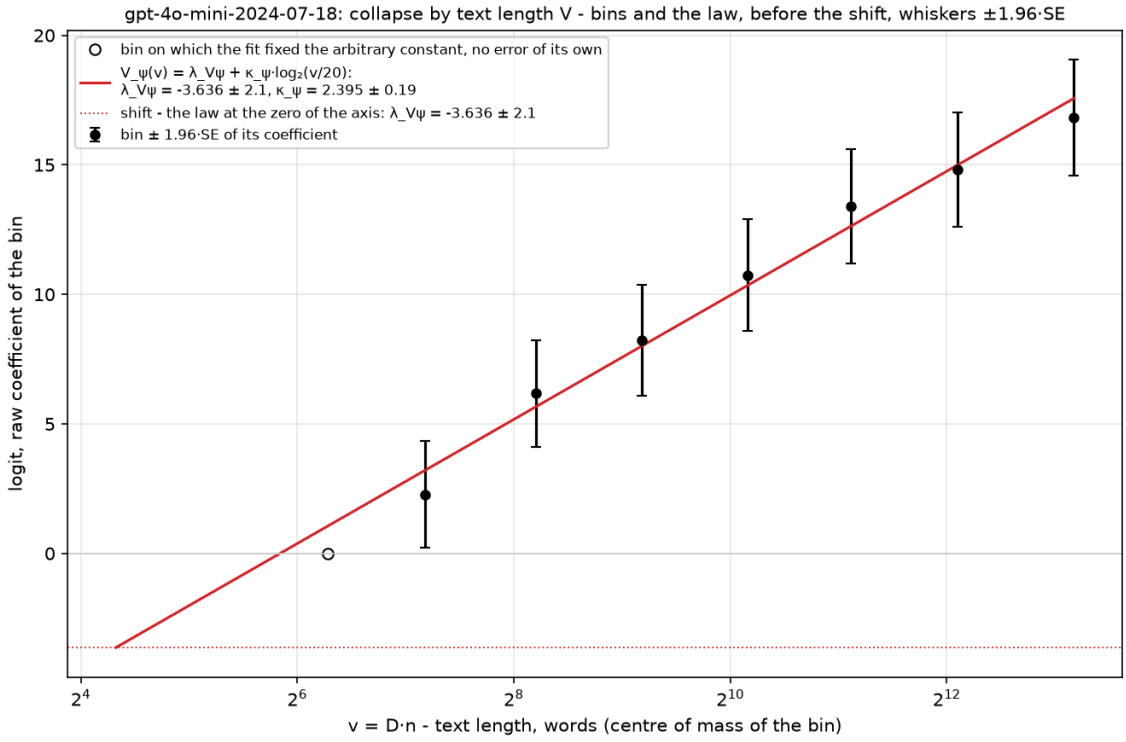


Figure 11: Collapse of the answer: text-length bins and their law, before the shift

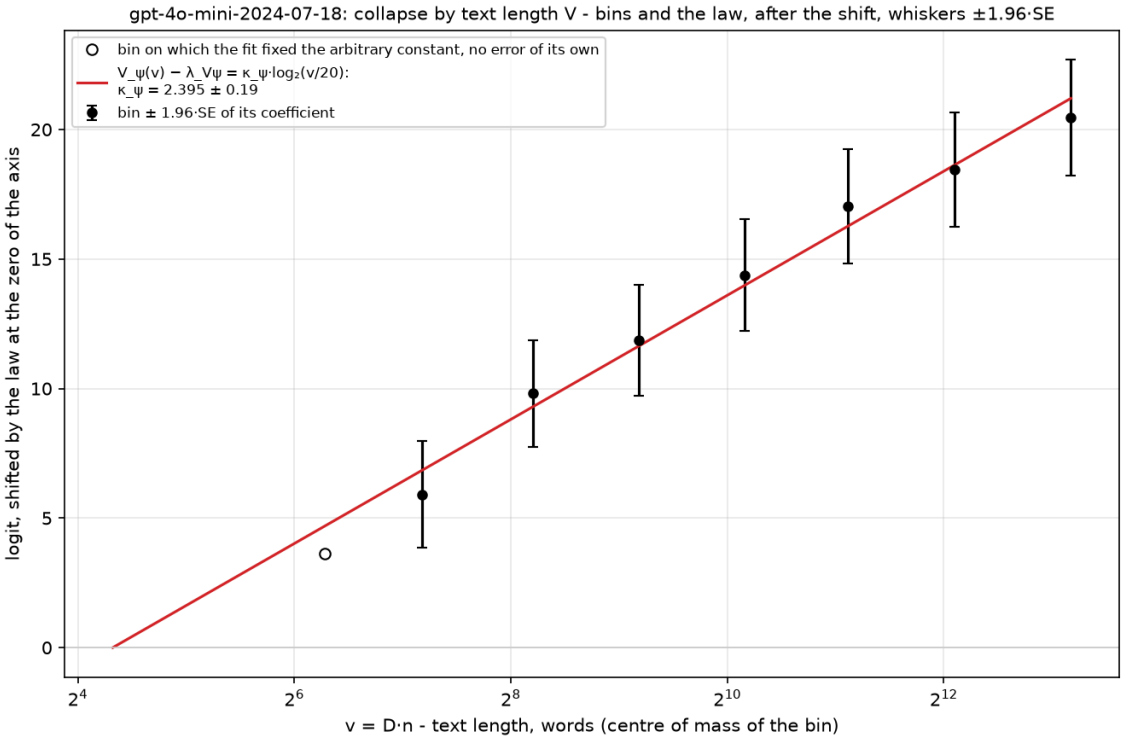


Figure 12: Collapse of the answer: text-length bins and their law, after the shift λ_Vψ

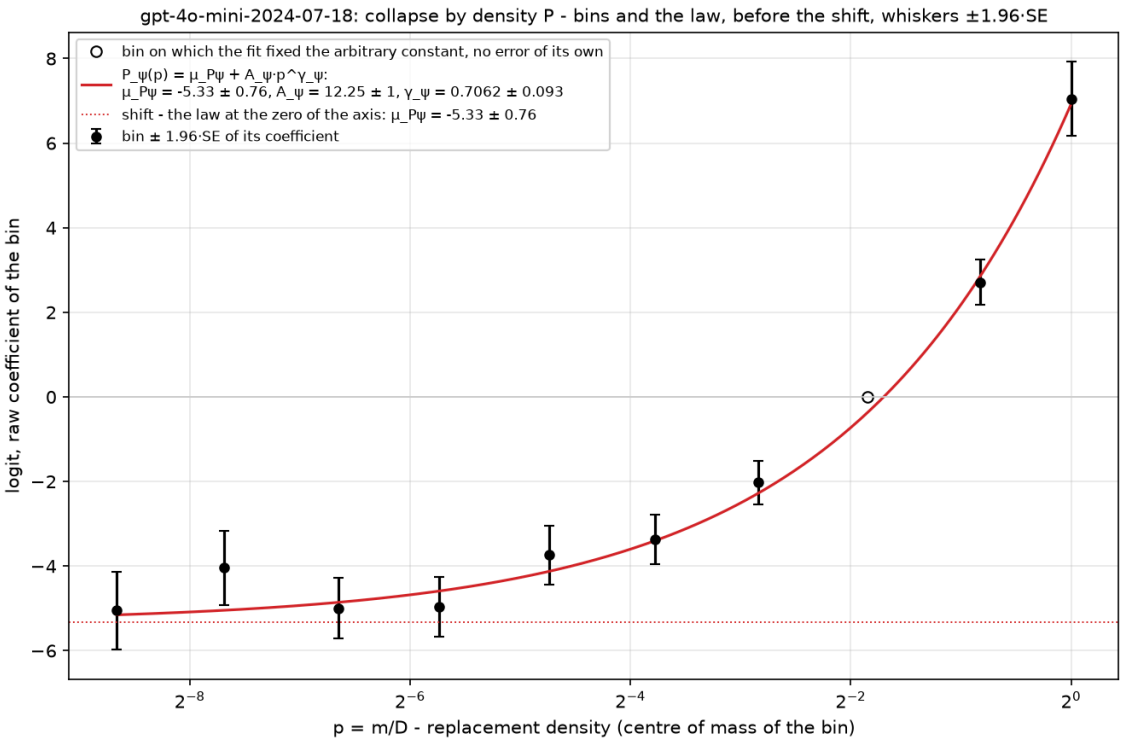


Figure 13: Collapse of the answer: density bins and their law, before the shift

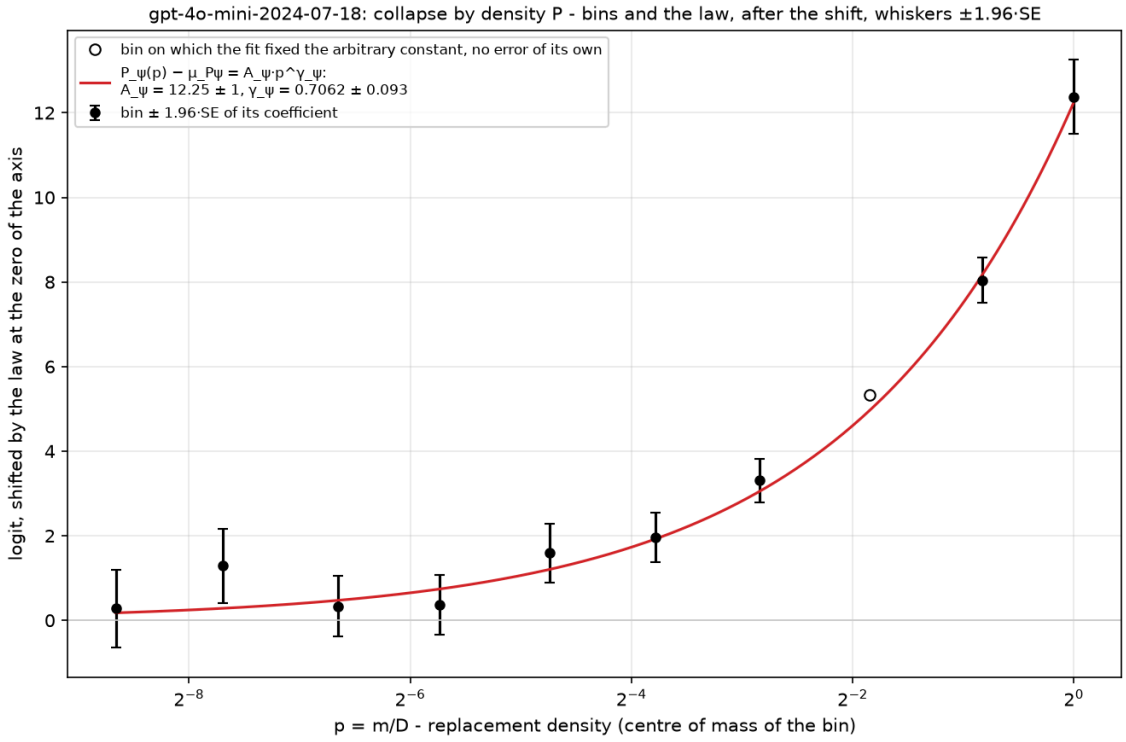


Figure 14: Collapse of the answer: density bins and their law, after the shift μ_Pψ

Place

The probability F that a point which lost the text has lost it by the word s. Every replacement word of the 995 points that lost the text is one trial - 365955 trials, 180097 of them on or after the line where the answer lost the text; the layout and the weights are those of these points. All laws are drawn over the full covariance of their bins:

$$\ln\frac{F}{1-F} = \alpha_F + S_F(\text{bin of } s) + P_F(\text{bin of } p) + Q_F\left(\text{ bin of } \frac{s-1}{v}\right)$$

$$S_F(s) = \mu_{SF} + \frac{\kappa_{SF}}{2}\left[\log_2\frac{s}{s_*} + \sqrt{\log_2^2\frac{s}{s_*} + \omega_{SF}^2}\right], \qquad P_F(p) = \mu_{PF} + A_F\,p^{\gamma_F}, \qquad Q_F = \lambda_{QF} + \kappa_{QF}\,\frac{s-1}{v}$$

$$\ln\frac{F}{1-F} = \lambda_F + \frac{\kappa_{SF}}{2}\left[\log_2\frac{s}{s_*} + \sqrt{\log_2^2\frac{s}{s_*} + \omega_{SF}^2}\right] + A_F\,p^{\gamma_F} + \kappa_{QF}\,\frac{s-1}{v}, \qquad \lambda_F = \alpha_F + \mu_{SF} + \mu_{PF} + \lambda_{QF}$$

- weighting by the cells of the run cube V × P × M: 161 cells, 995 points, largest occupancy 41; effective points n_eff = 475.68

Parameter	Value ± 1.96·SE	Note
λ_F	−10.65 ± 1.99	logit of the place on the plateau, at p ⟶ 0, at the start
κ_SF	1.14 ± 0.209	rise past the corner, logit per doubling of the word
s_*	39.57 ± 22.63	corner: the word number where the asymptotes cross
ω_SF	0.00100 (held on its bound)	width of the corner, doublings of the word number
A_F	6.95 ± 1.56	density term at the full density p = 1
γ_F	0.420 ± 0.155	exponent of the density term
κ_QF	3.19 ± 0.589	rise over the whole text, logit
μ_SF	−5.15 ± 1.37	word shift: the plateau of the word law, s ⟶ 0
μ_PF	−3.99 ± 1.53	density shift: the density law at p ⟶ 0
λ_QF	−0.130 ± 0.640	share shift: the share line at the start of the text
χ^2 / points, `place_word`	4.53 / 10	the law against its bins
χ^2 / points, `place_density`	9.09 / 10	the law against its bins
χ^2 / points, `place_share`	42.91 / 9	the law against its bins

The bins are fitted on 365955 trials (180097 with the text lost); a bin whose trials all have one outcome has no estimate and goes out with its trials: none. The coefficients are raw logits relative to the reference bin of the axis (`0`), before the shift.

Axis	Bin	Value ± 1.96·SE
word number s	1–10	−5.02 ± 1.71
word number s	11–20	−5.08 ± 1.52
word number s	21–40	−5.16 ± 1.48
word number s	41–80	−4.70 ± 0.988
word number s	81–160	−3.17 ± 0.881
word number s	161–320	−2.21 ± 0.646
word number s	321–640	−0.951 ± 0.411
word number s	641–1280	0.00
word number s	1281–2560	1.11 ± 0.434
word number s	2561–5120	2.42 ± 0.850
word number s	5121–10240	3.15 ± 1.34
replacement density p	1/1024–1/512	−3.12 ± 2.51
replacement density p	1/512–1/256	−4.40 ± 1.33
replacement density p	1/256–1/128	−3.26 ± 1.14
replacement density p	1/128–1/64	−3.17 ± 1.04
replacement density p	1/64–1/32	−3.47 ± 1.19
replacement density p	1/32–1/16	−2.83 ± 1.08
replacement density p	1/16–1/8	−2.01 ± 0.947
replacement density p	1/8–1/4	−1.09 ± 0.796
replacement density p	1/4–1/2	0.00
replacement density p	1/2–1	1.64 ± 0.696
replacement density p	1	2.84 ± 0.830
share of the text (s − 1)/v	0–0.1	0.00
share of the text (s − 1)/v	0.1–0.2	0.311 ± 0.654
share of the text (s − 1)/v	0.2–0.3	0.882 ± 0.783
share of the text (s − 1)/v	0.3–0.4	1.16 ± 0.831
share of the text (s − 1)/v	0.4–0.5	1.65 ± 0.836
share of the text (s − 1)/v	0.5–0.6	1.27 ± 0.918
share of the text (s − 1)/v	0.6–0.7	1.68 ± 0.924
share of the text (s − 1)/v	0.7–0.8	2.02 ± 0.912
share of the text (s − 1)/v	0.8–0.9	2.42 ± 0.905
share of the text (s − 1)/v	0.9–1	3.21 ± 0.907

Check, the text length: the bins of the text length fitted together with the bins above; Wald statistic «all of them are zero» 13.89 with 7 degrees of freedom, p = 0.0531; the largest change of a bin above 2.32 of its SE.

Text length	Value ± 1.96·SE
41–80	0.00
81–160	−0.155 ± 0.902
161–320	−0.132 ± 1.13
321–640	−0.645 ± 1.34
641–1280	−0.112 ± 1.55
1281–2560	0.171 ± 1.66
2561–5120	0.669 ± 1.87
5121–10240	1.62 ± 2.18

Check, the number of rules: the bins of the number of rules fitted together with the bins above; Wald statistic «all of them are zero» 27.74 with 8 degrees of freedom, p = 0.000526; the largest change of a bin above 1.06 of its SE.

Number of rules	Value ± 1.96·SE
1	0.00
2	−0.902 ± 0.995
3–4	−1.10 ± 0.761
5–8	−0.497 ± 0.688
9–16	−1.09 ± 0.732
17–32	−1.44 ± 0.768
33–64	−1.15 ± 0.857
65–128	−1.90 ± 0.961
129–256	−0.313 ± 1.62

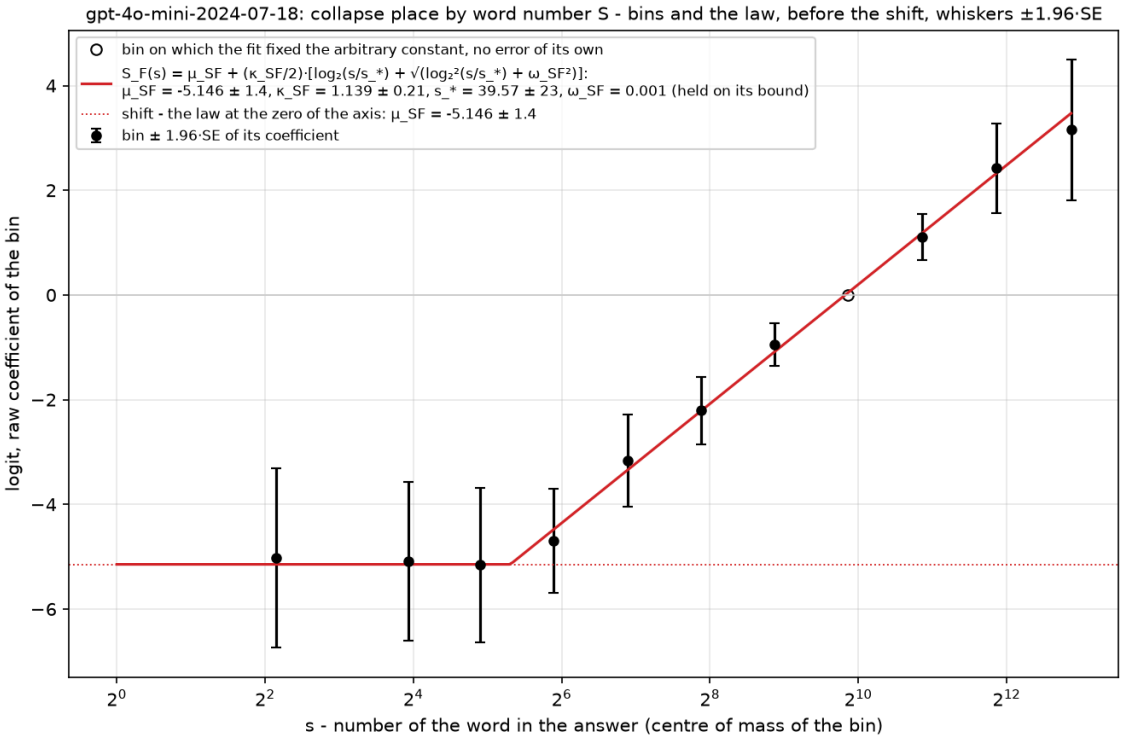


Figure 15: Place of the collapse: word-number bins and their law, before the shift

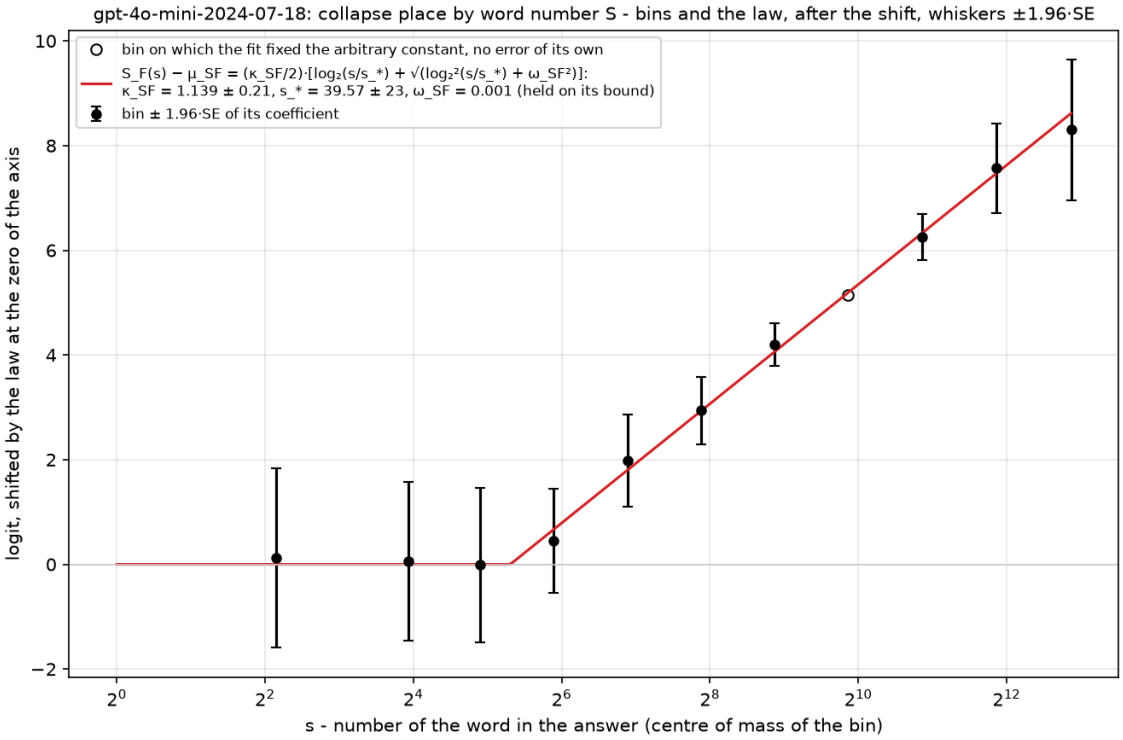


Figure 16: Place of the collapse: word-number bins and their law, after the shift μ_SF

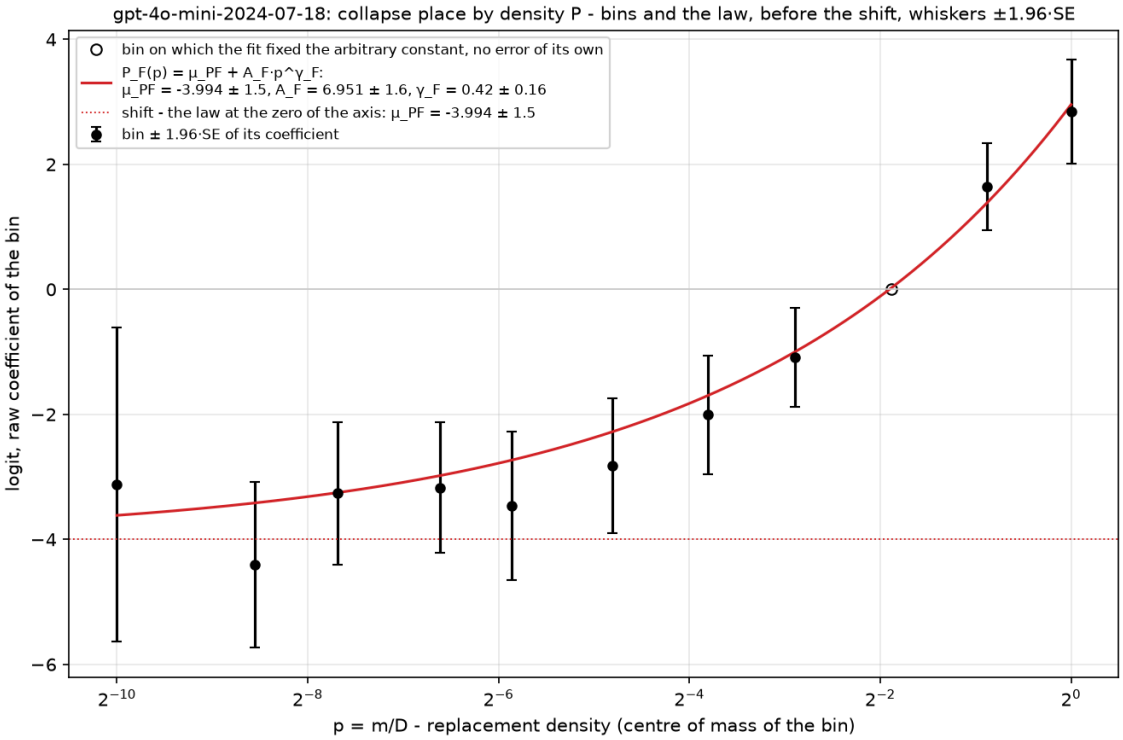


Figure 17: Place of the collapse: density bins and their law, before the shift

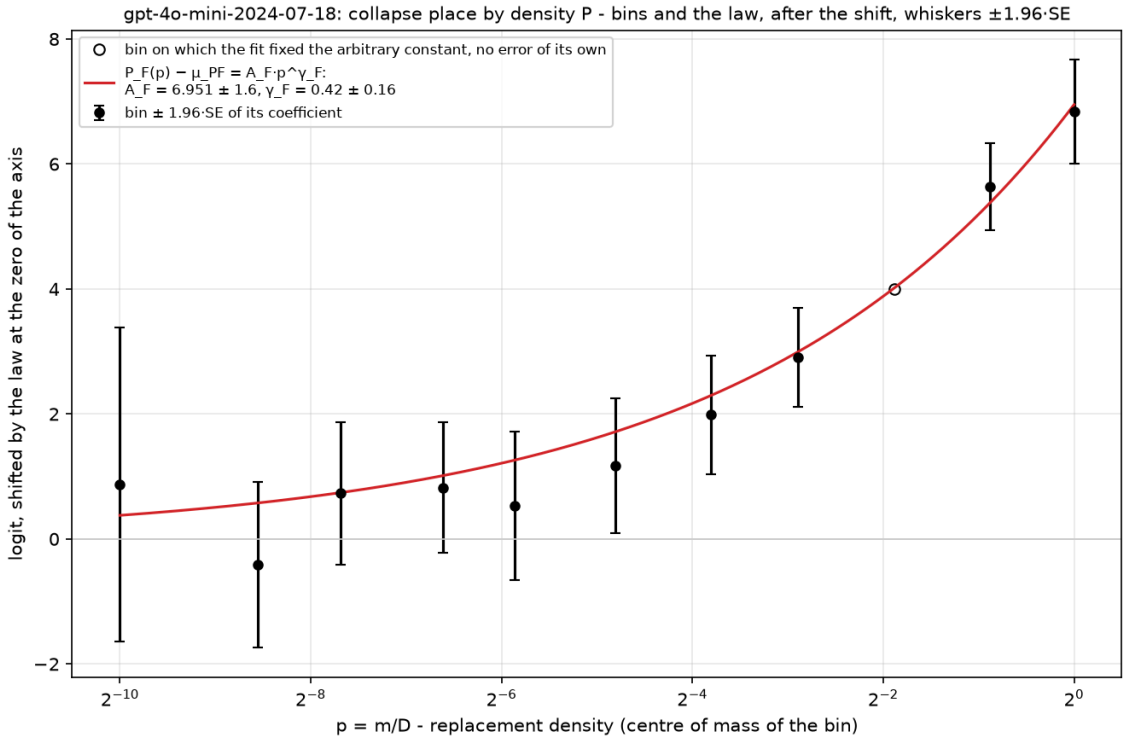


Figure 18: Place of the collapse: density bins and their law, after the shift μ_PF

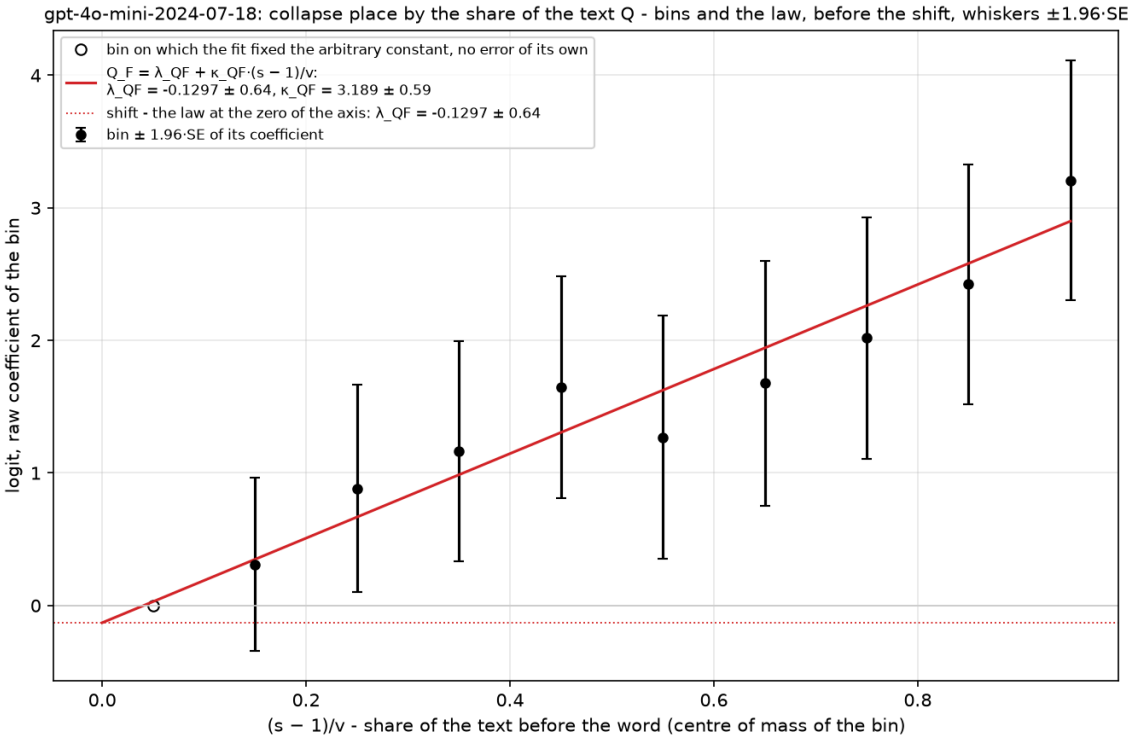


Figure 19: Place of the collapse: bins of the share of the text and their line, before the shift

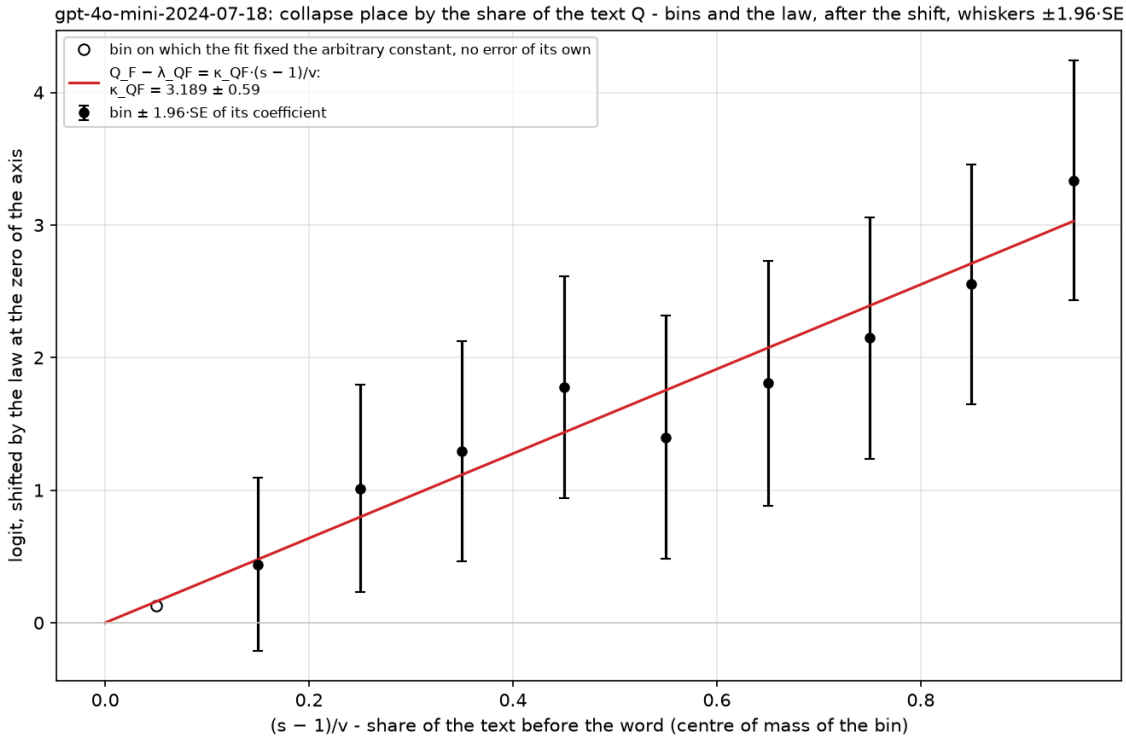


Figure 20: Place of the collapse: bins of the share of the text and their line, after the shift λ_QF

Bin cube filling

`bin cube: point counts by bins D × m × n`

`m ∈ 1`

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	3	3	3	3	3	9	39	3
5-8	3	3	3	3	6	8	3	3
9-16	3	3	3	3	3	29	27	50
17-32	3	3	3	3	3	11	3	x
33-64	3	3	3	5	6	15	x	x
65-128	3	6	6	10	3	x	x	x
129-256	3	22	24	3	x	x	x	x
257-512	3	7	11	x	x	x	x	x
513-1024	3	3	x	x	x	x	x	x

`m ∈ 2`

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	4	4	4	27	6	37	3	x
5-8	3	3	3	3	3	9	3	x
9-16	3	3	3	3	8	3	3	3
17-32	3	3	9	6	10	12	3	x
33-64	3	3	3	19	18	3	x	x
65-128	3	3	3	14	16	x	x	x

129-256	3	3	24	44	x	x	x	x
257-512	3	14	3	x	x	x	x	x
513-1024	7	12	x	x	x	x	x	x

m ∈ 3-4

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	3	3	11	50	3	3	x	x
5-8	3	3	6	6	3	3	x	x
9-16	4	3	5	5	3	x	x	x
17-32	5	8	5	7	28	3	x	x
33-64	6	3	3	11	37	3	x	x
65-128	6	3	14	26	3	x	x	x
129-256	6	3	3	3	x	x	x	x
257-512	15	14	15	x	x	x	x	x
513-1024	41	26	x	x	x	x	x	x

m ∈ 5-8

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	x	x	x	x	x	x	x	x
5-8	6	18	46	3	x	x	x	x
9-16	6	29	46	68	3	x	x	x
17-32	19	20	11	14	40	3	x	x
33-64	6	3	6	5	75	3	x	x
65-128	6	10	12	40	3	x	x	x
129-256	15	17	11	3	x	x	x	x
257-512	40	23	37	x	x	x	x	x
513-1024	3	3	x	x	x	x	x	x

m ∈ 9-16

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	x	x	x	x	x	x	x	x
5-8	x	x	x	x	x	x	x	x
9-16	44	20	3	3	x	x	x	x
17-32	8	26	23	3	3	x	x	x
33-64	6	5	9	3	3	x	x	x
65-128	9	6	25	3	x	x	x	x
129-256	12	15	3	x	x	x	x	x
257-512	21	22	3	x	x	x	x	x
513-1024	17	3	x	x	x	x	x	x

m ∈ 17-32

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	x	x	x	x	x	x	x	x
5-8	x	x	x	x	x	x	x	x
9-16	x	x	x	x	x	x	x	x
17-32	91	3	3	x	x	x	x	x
33-64	29	31	3	x	x	x	x	x
65-128	17	12	42	3	x	x	x	x
129-256	11	20	50	3	x	x	x	x

257-512	25	42	3	x	x	x	x	x
513-1024	3	3	x	x	x	x	x	x

m ∈ 33-64

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	x	x	x	x	x	x	x	x
5-8	x	x	x	x	x	x	x	x
9-16	x	x	x	x	x	x	x	x
17-32	x	x	x	x	x	x	x	x
33-64	63	3	x	x	x	x	x	x
65-128	76	61	3	x	x	x	x	x
129-256	32	3	3	x	x	x	x	x
257-512	3	3	x	x	x	x	x	x
513-1024	x	x	x	x	x	x	x	x

m ∈ 65-128

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	x	x	x	x	x	x	x	x
5-8	x	x	x	x	x	x	x	x
9-16	x	x	x	x	x	x	x	x
17-32	x	x	x	x	x	x	x	x
33-64	x	x	x	x	x	x	x	x
65-128	3	3	x	x	x	x	x	x
129-256	73	3	x	x	x	x	x	x
257-512	74	3	x	x	x	x	x	x
513-1024	3	x	x	x	x	x	x	x

m ∈ 129-256

D\n	1-5	6-10	11-20	21-40	41-80	81-160	161-320	321-640
1-4	x	x	x	x	x	x	x	x
5-8	x	x	x	x	x	x	x	x
9-16	x	x	x	x	x	x	x	x
17-32	x	x	x	x	x	x	x	x
33-64	x	x	x	x	x	x	x	x
65-128	x	x	x	x	x	x	x	x
129-256	3	x	x	x	x	x	x	x
257-512	3	x	x	x	x	x	x	x
513-1024	3	x	x	x	x	x	x	x